

Conjugated Systems & Aromaticity

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1 Conjugated Systems

1.1 Organic Acids & Bases Review

Core Concept 1.1: General Organic Compound Acidity

Memorizing the following pK_a values as baseline references is essential—they serve as anchors for estimating the acidities of related compounds by comparing functional groups, electron-withdrawing effects, and resonance stabilization of the conjugate base.

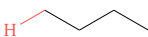
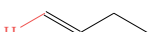
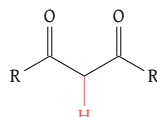
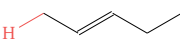
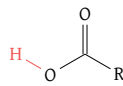
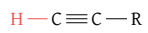
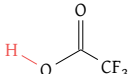
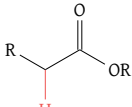
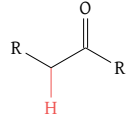
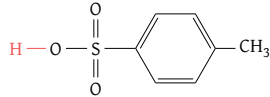
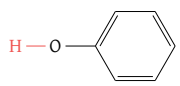
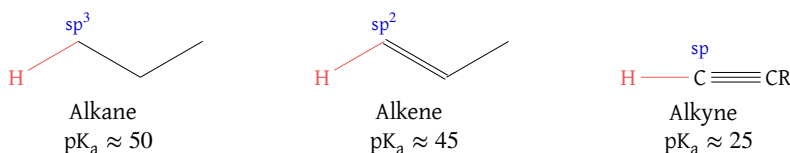
Compound	pK_a	Structure	Compound	pK_a	Structure
Alkane	50		Ammonium	10	
Alkene	45		α to Two Carbonyls	10	
Allylic	40		Carboxylic Acid	5	
Amine	35		Oxonium	0	
Alkyne	25		Trifluoro Acetic Acid	0	
α to an Ester	25		Hydrohalic Acid	<0	
α to a Ketone or Aldehyde	20		Sulfuric Acid	<0	
Alcohol (ROH)	15		Toluenesulfonic Acid	<0	
Phenol (PhOH)	10				

Table 1: Common acids and their pK_a values, ordered from least to most acidic

Core Concept 1.2: Acidity of Alkanes, Alkenes, and Alkynes

We now examine the pK_a values trend of alkanes, alkenes, and alkynes. The linear pattern observed can be rationalized by considering the following factors.

Figure 1: Alkane, alkene, and alkyne pK_a values

- (a) **Electronegativity of atom attached to the acidic H:** This factor cannot account for the trend, since the acidic hydrogen is attached to a carbon atom in all three cases, precluding any difference in electronegativity.
- (b) **Inductive Effect:** This factor is likewise insufficient, as only C–H bonds are present in the vicinity of the acidic proton.
- (c) **Resonance:** No resonance stabilization of the conjugate base is possible, so this factor does not explain the trend.
- (d) **Charge:** All species are neutral, so this factor offers no explanation.
- (e) **Hybridization:** Here a genuine difference emerges, as the acidic carbon is sp^3 -, sp^2 -, or sp -hybridized in the alkane, alkene, and alkyne, respectively. The lone pair generated upon deprotonation is stabilized most effectively in the sp -hybridized orbital, since its greater s character holds the electrons closer to the nucleus.

When the acidity cannot be rationalized from the structure of the acid itself, it is useful to instead examine the **conjugate bases**.

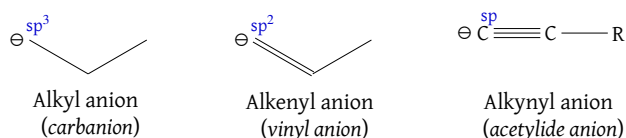
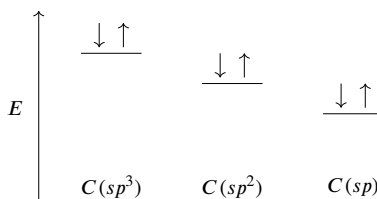


Figure 2: Conjugate bases of an alkane (left), alkene (middle), and alkyne (right)

Among the anions shown in Figure 2, the **alkynyl anion is the weakest base** because its lone pair is the most stabilized and thus least reactive. This arises because its e^- density is drawn more toward the carbon nucleus—away from the bonding region—making it less attractive toward H^+ relative to the alkenyl and alkyl anions. The following energy diagram below illustrates this comparison.

Figure 3: Diagram of sp^3 , sp^2 , and sp orbital energy levels; lone pair in the sp orbital is most stabilized because the relatively greater s character increases the probability e^- 's residing closer to the positive charge of the nucleus

Core Concept 1.3: K_{eq} and pK_a s

Any acid–base reaction has an equilibrium constant given by $K_{\text{eq}} = 10^{(\text{pK}_{a,\text{conj. acid}} - \text{pK}_{a,\text{acid}})}$. Using the pK_a values in Table 1, we can therefore estimate the K_{eq} of a given acid–base reaction to a reasonable degree of accuracy, as illustrated in the following examples.

(a) Protonation of ammonia by hydronium

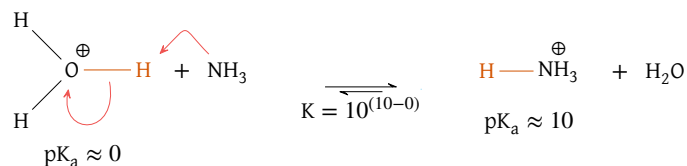


Figure 4: Protonation of ammonia by hydronium

(b) Enolate formation

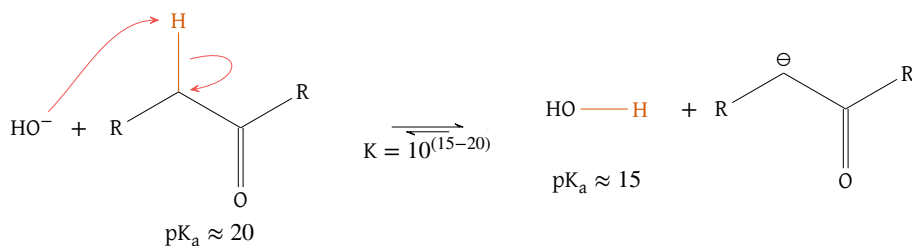


Figure 5: Enolate formation

(c) C-protonation of an enolate

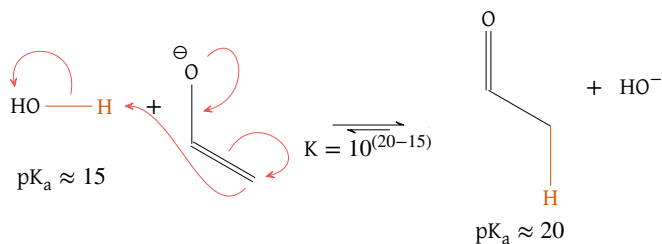


Figure 6: C-protonation of an enolate

(d) Enol formation

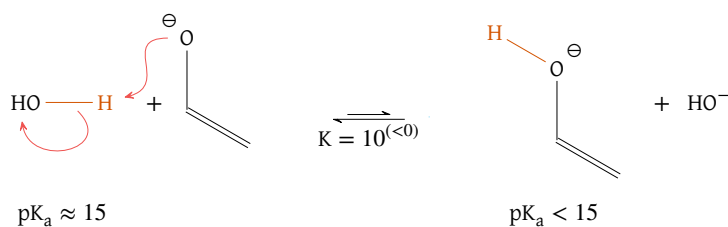
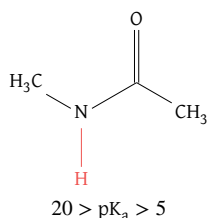


Figure 7: Enol formation

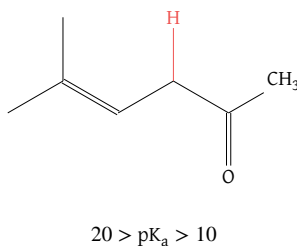
Furthermore, pK_a values of related structures can be predicted from the reference values in Table 1, as demonstrated in the following examples.

(a) Amine like α to carbonyl

Figure 8: Amine like α to carbonyl

The above molecule has a higher pK_a than a carboxylic acid ($pK_a = 5$), since nitrogen is less electronegative than oxygen, but a lower pK_a than a proton α to a carbonyl group ($pK_a = 20$), since nitrogen is more electronegative than carbon.

(b) α to carbonyl and allylic

Figure 9: α to carbonyl group and allylic

Here, The above molecule has a higher pK_a than a proton α to two carbonyl groups ($pK_a = 10$), but a lower pK_a than one α to a single carbonyl group ($pK_a = 20$), since it is additionally allylic.

1.2 Allylic Systems

Core Concept 1.4: Allylic Carbons

Allylic carbons are sp^3 hybridized carbons directly attached to a $C=C$ double bond, and an **allylic hydrogen** is one bonded to such a carbon.

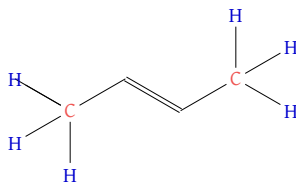


Figure 10: Allylic carbons (red) and allylic hydrogens (blue)

Core Concept 1.5: Allylic System Resonance

An allylic hydrogen exhibits greater acidity than both alkane and alkyne hydrogens, as resonance delocalizes the negative charge across the resulting allylic anion, thereby rendering the anion a comparatively weaker base.

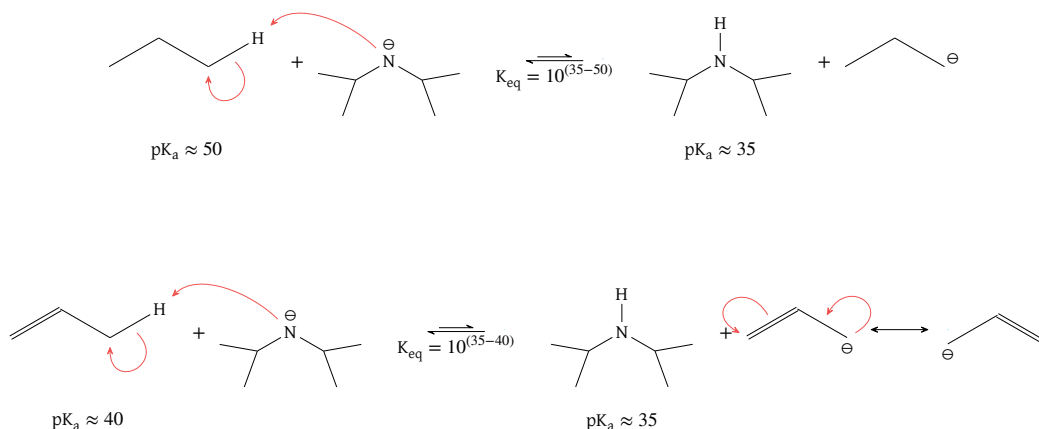


Figure 11: K_{eq} s of alkane (top) vs allylic (bottom) hydrogens; the K_{eq} of the allylic scenario is much more forward favoring because it is able to better stabilize the negative anion state charge

Just as negative charge (filled octets) is stabilized by resonance within allylic structures, radicals and positive charge (unfilled octets) are stabilized through this same resonance mechanism.

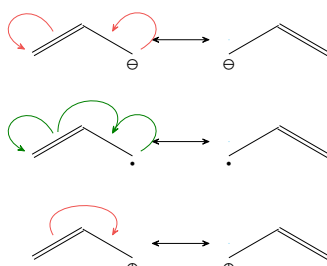


Figure 12: Filled octets (top), radicals (middle), and unfilled octets (bottom)

Critically, **any lone pair capable of participating in resonance must reside in a p orbital.**

Core Concept 1.6: Allylic System Bond Dissociation Energy

As a consequence of resonance-induced stabilization, the bond dissociation energies of distinct hydrogens within an alkene follow a predictable trend.

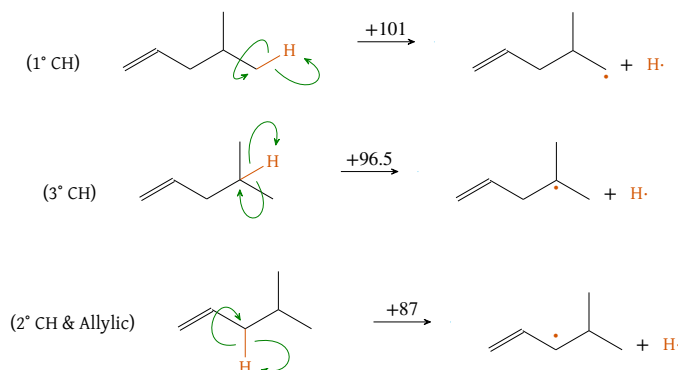


Figure 13: Primary (top), tertiary (middle), and secondary & allylic (bottom) hydrogen bond dissociation energies within the same alkene; secondary allylic hydrogen bond takes least energy to break due to resonance-induced stabilization

Above, the 3° carbon radical is lower in energy than the 1° carbon radical as a result of hyperconjugation between the radical carbon and neighboring σ bonds. The allylic radical, by contrast, receives comparatively less hyperconjugative stabilization as a 2° carbon, yet is stabilized through resonance, causing its corresponding bond dissociation to require the least energy of the three.

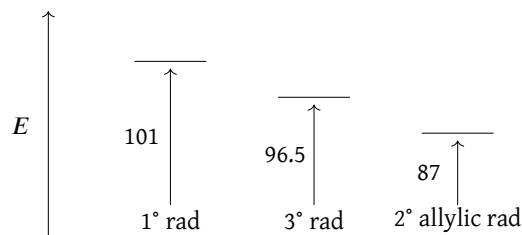


Figure 14: Energy diagram of Figure 13

This differential stabilization gives rise to chemical consequences that play a substantial role in practice, as illustrated by the reaction below.

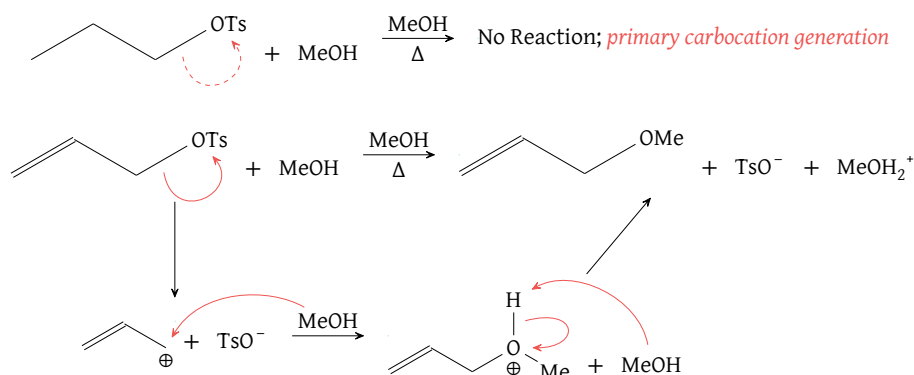


Figure 15: Non-allylic (top) vs allylic (bottom) leaving group primary carbocation formation viability; a primary carbocation can form in the bottom reaction because the cation is additionally allylic; Ts stands for a tosyl group, a common chemical group used in organic synthesis

Core Concept 1.7: Allylic System Molecular Orbital Structure

The molecular orbitals **relevant to allylic structures are the p orbitals**, whose energy levels can be compared in the context of sp - and sp^2 -hybridized carbons, as shown below. Notably, the energy level of sp^2 orbitals lies higher than that of sp orbitals.



Figure 16: Orbital energy level diagrams for sp and sp^2 hybridization

Superimposed on a skeletal diagram backbone, this orbital structure appears as follows.

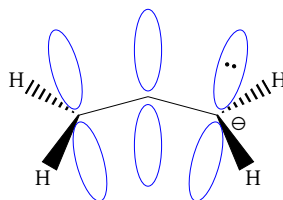


Figure 17: σ bond orbitals overlap along internuclear axis (handshake) whereas π bond orbitals overlap above and below internuclear axes. All carbons are sp^2 hybridized

Considered as a whole, the allylic p-orbital system produces an energy diagram with a distinctive three-level pattern, progressing from bonding to non-bonding to anti-bonding.

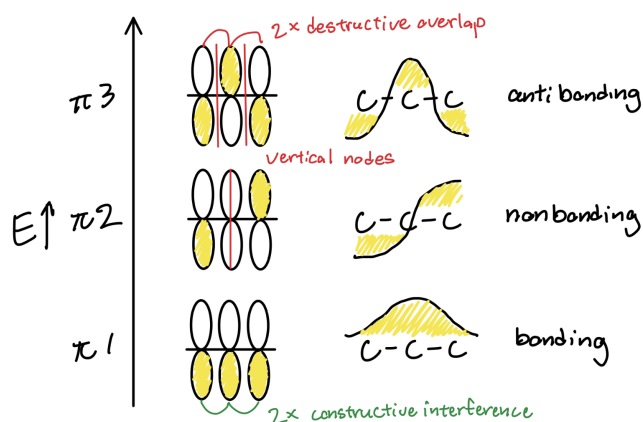


Figure 18: Energy diagram of allylic p-orbital system

Consequently, the HOMO and LUMO assignments differ among the allylic anion, radical, and cation.

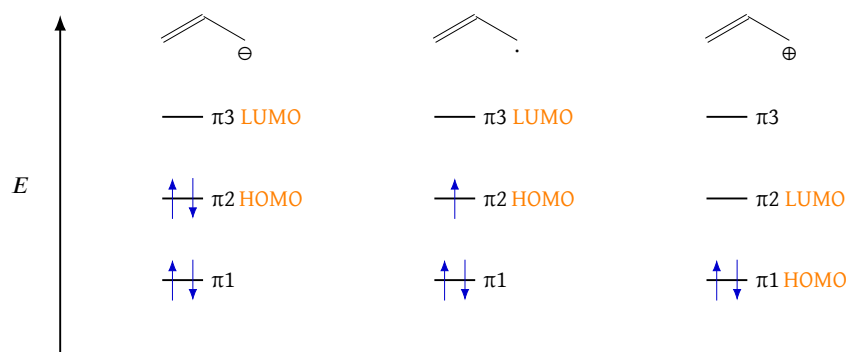


Figure 19: Molecular orbital diagrams for the allyl anion, radical, and cation, showing the filling of π_1 , π_2 , and π_3 with 4, 3, and 2 πe^- s respectively, along with HOMO LUMO assignments

This distinction is particularly relevant in the context of **frontier molecular orbital theory**, which holds that the HOMO of a nucleophile interacts with the LUMO of an electrophile (or vice versa) as explored in the context of diel alder reaction in UNCOMMENT AT END

1.3 Conjugated π Systems

Core Concept 1.8: Conjugated π System Stability

A conjugated pi system refers to network of adjacent allylic structures that allow for resonance across a greater number of carbons. For instance, take a look at the hydrogenation enthalpies of the following alkenes.

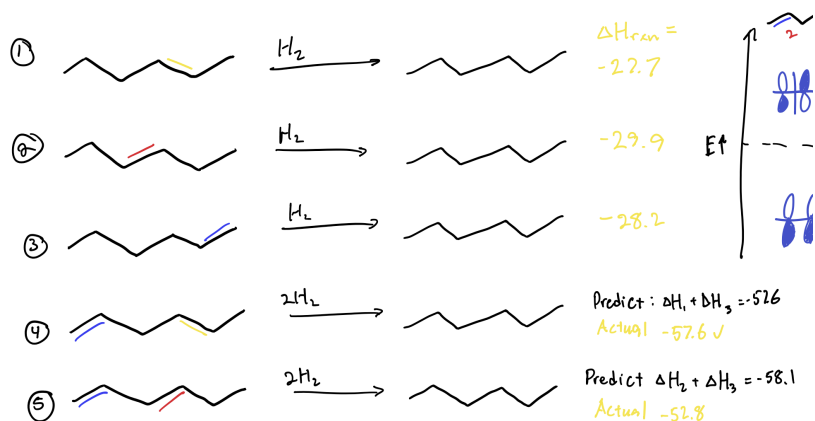


Figure 20

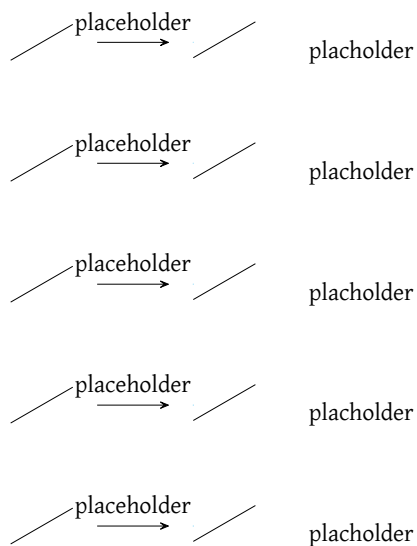


Figure 21: caption

Compound 5 is more stabilized than predicted based on intuitive addition of constituent enthalpies because of the ability of conjugated π systems to better stabilize excess or lacking charge.

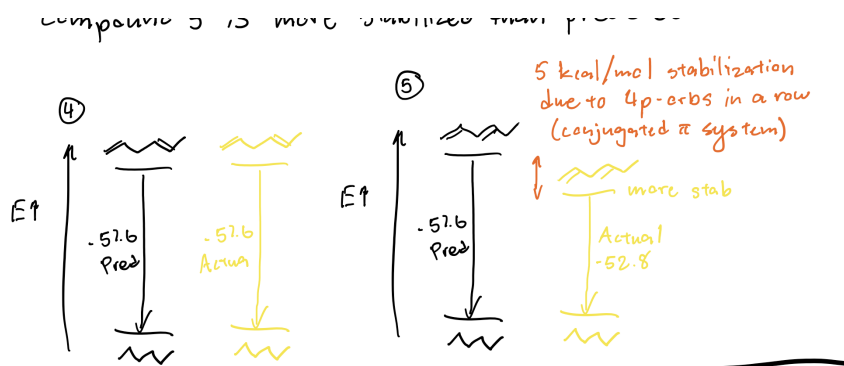
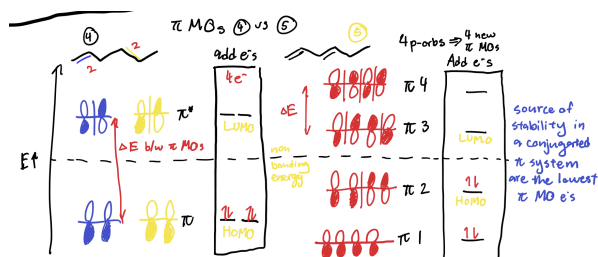


Figure 22

When we look closer into the orbital energy diagrams of these conjugated π systems like we have done in REFERENCE TO 3 PI SYSTEM, we find that the conjugated π system orbital energy levels sit in an overall larger range but possesses smaller stepwise increases in energy, thus the HOMO LUMO gap is comparatively smaller compared to the non conjugated π system alkenes.

Figure 23: source of stability in a conjugated π system are the lowest π MO e^- s

The rules are the following.

- As conjugation increases (more p orbitals in a row), the energy gap between π MOs get smaller
- As conjugation increases, the lowest energy π MO gets lower in energy and the highest energy π MO goes higher in energy

It's important to know that every high energy intermediate needs to be as stabilized as possible and that resonance from conjugated π systems trump stability originating from hyperconjugation.

$$\text{resonance} > 3^\circ > 2^\circ > 1^\circ$$

This has particularly large consequences in the different reactions covered within Alkenes & Alkynes because many electrophilic additions progress in a Markovnikov or anti-Markovnikov manner, which takes into consideration intermediate state stability. Examples include alkene hydrohalogenation (Core Concept 1.9), which progresses in a Markovnikov manner, and hydroboration-oxidation (Core Concept 1.13), which progresses in an anti-Markovnikov manner.

Using our learned knowledge of conjugated π system stability, let's look at two examples.

Core Concept 1.9: Chlorination of Conjugated Cycloalkene

Our goal is to predict all of the products of the following reaction, including stereoisomers.

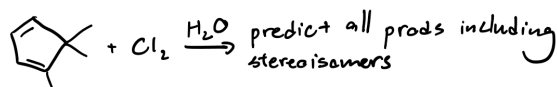


Figure 24

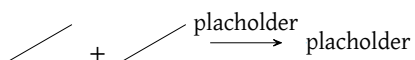


Figure 25: caption

First, we map out the chloronium intermediates.

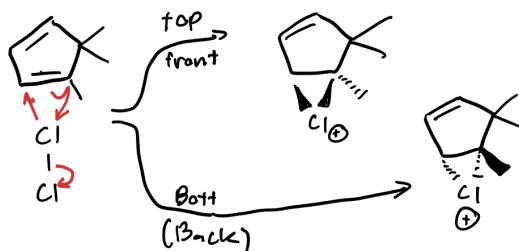


Figure 26

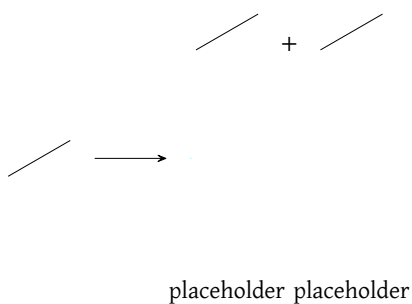


Figure 27: caption

Use resonance to determine which carbon of halonium to attack with weak nucleophilic ROH.

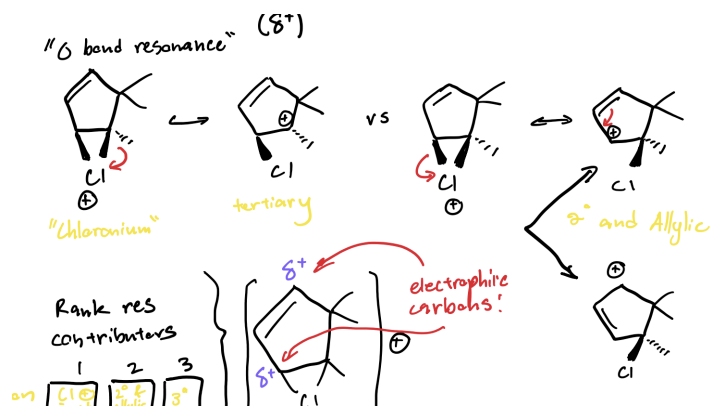


Figure 28: tertiary "σ bond resonance" vs 2° and allylic



Figure 29: caption

Because the allylic side is more stable, the electrophilic carbons exist in two locations within the system

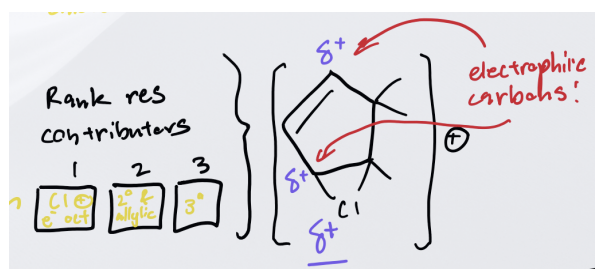


Figure 30: the top resonance contributor is the Cl

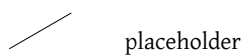


Figure 31: caption

Thus three possible attacks exist, giving us three distinct stereoisomeric products.

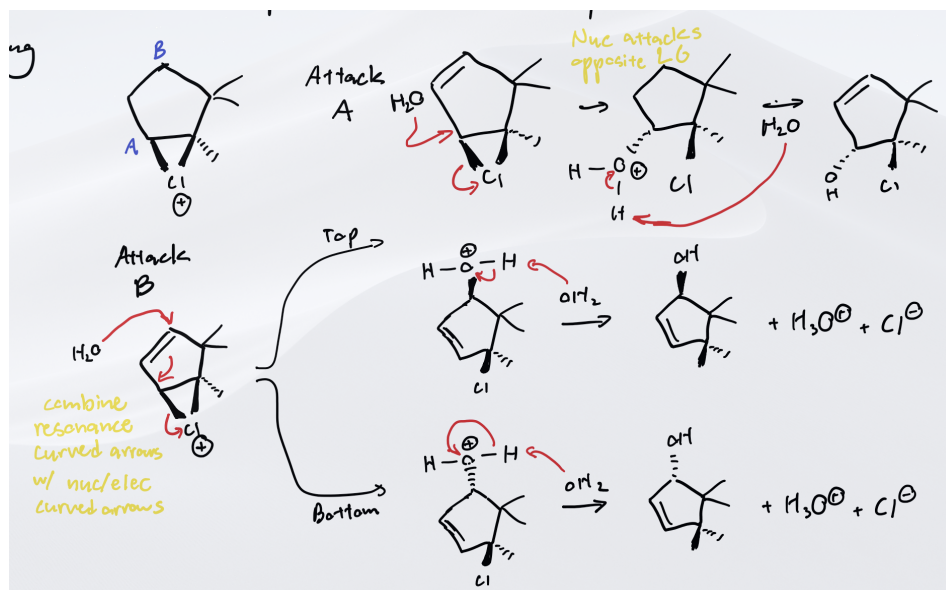


Figure 32: use this as the reference for three separate figures below

(a) attack name placeholder



Figure 33: caption

(b) attack name placeholder



Figure 34: caption

(c) attack name placeholder

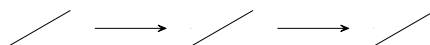


Figure 35: caption

Core Concept 1.10: Radical Bromination of Terminal Alkene

The radical bromination of a terminal alkene works in a similar manner to the reaction shown in Core Concept 1.9, wherein resonance creates the possibility of multiple different types of attacks and consequently three distinct products.

We first create Br_2 through N-Bromo-succinimide, which allows for the production of **monobrominated products**. Realize that adding pure Br_2 to react with alkenes results in dihalogenation as mapped out in Core Concept 1.11.

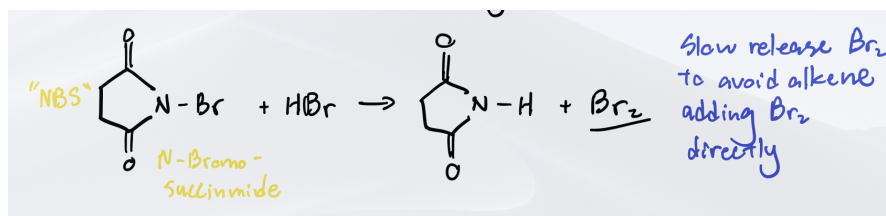
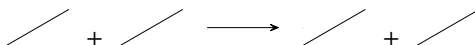
Figure 36: slow release of Br₂ to avoid alkene adding Br₂ directly

Figure 37: caption

After the spontaneous dissociation of Br₂ into radical bromine atoms, the first propagation occurs, exposing a radical in the original alkene compound.

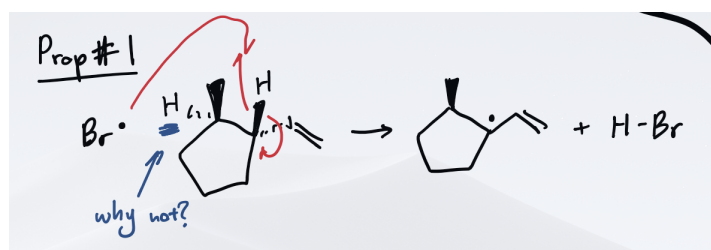


Figure 38



Figure 39: caption

This radical alkene and the remaining bromine radical can carry out a second propagation interaction in three different manners.

(a) **propagation placeholder**

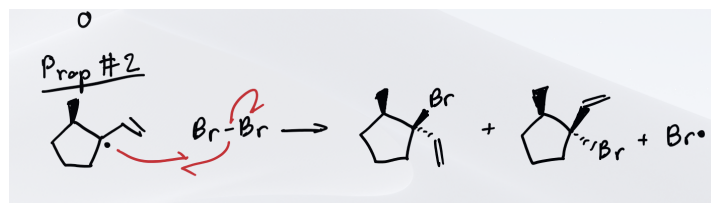


Figure 40



Figure 41: caption

(b) propagation placeholder

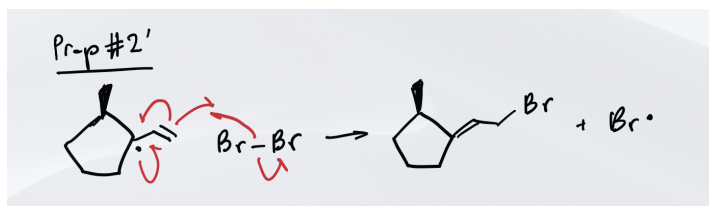


Figure 42

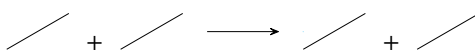


Figure 43: caption

(c) propagation placeholder

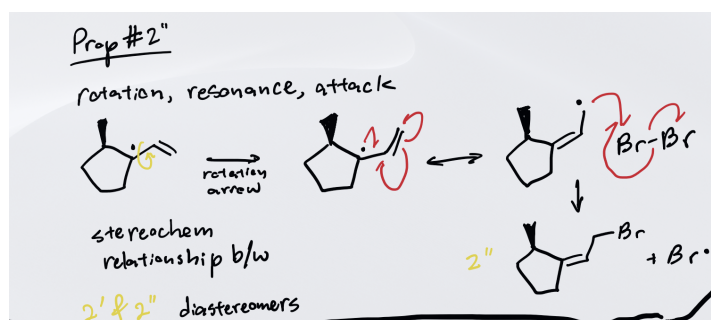


Figure 44



Figure 45: caption

To finalize, the full reaction looks like this.

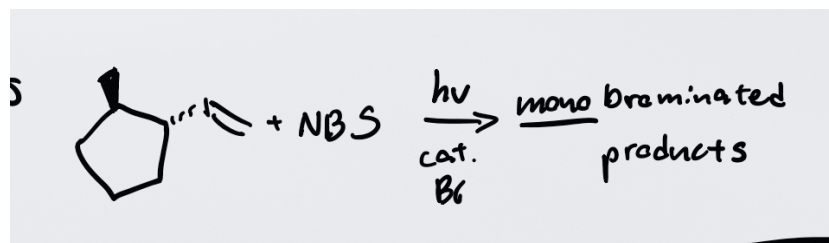


Figure 46



Figure 47: caption